Unit – VI

VLSI Testing and Analysis

Topics

- Fault, Need for Testability, DFT Guidelines, Testability, fault models, Path Sensitizing,
- Test pattern gener^n, Seque. Circuit Test, Built in Self Test.
- JTAG & Boundary Scan & TAP Controller.

Faults

- The faults means a certain error is generated at the o/p
- This error may be Develop Durning manifacturing of ckt.

* There are three main Types of faults.

① Stuck-At fault [logical]
② Bridging fault [shorts]
③ Delay faults [Timing]
④ Transistor faults.

① Stuck-At faults.

- stuck at-zero
- A Particular node in the Circuit permanently remains at logic zero
- That means permanently stuck at logic zero.
- most Common & Simplest fault.

ⓐ Stuck-at-0 → The signal line is permanently fixed at a logic '0' [GND] Cannot change to '1'

ⓑ Stuck at-1 →
- A particular Node in the ckt Permanently remains at logic one
- Node is permanently stuck at one

② Bridging fault. [shorts].

- when two or more conducting path get short circuited.
- Then it produces error in o/p is called Bridging fault.
- It causes incorrect logic and even lead to the circuit oscillating.

② Delay Faults - [Timing].

- chip's logic is correct, but the signal is too slow.

③ Types of delay

① Great delay
- It is the time required when signal Propagating from logical gate.

② Path delay.
- It depends on path followed by signal to reach the o/p.

④ Transister faults

① Stuck open ckt
- A transister is Permanently off open., Vtg is on its gate.
- Can turn a logic gate into a Seq. ckt, which is diffcult to test

② Stuck-Short fault

- A transister is permanently on Shorted, Vtg is on its gate.

Design for Testability

- DFT is a Technique used to make its easier to test after manufacturing to check they work correctly
- and find defects.

* Need of DFT

① IC Testing is time Consumping & expensive DFT makes it simple

② Makes Testing easy even for veey Complex IC's with millions of Component

③ Saves Time because testing is automatic.

④ Detects Common faults easily, like Stuck at-0, stuck at-1, Timing and bridging faults.

⑤ Impeoves yield

⑥ Improve reliability by finding faults eaely.

⑦ Improve Support BIST for automatic

testing inside the chip.

⑧ Improve Test Coverage

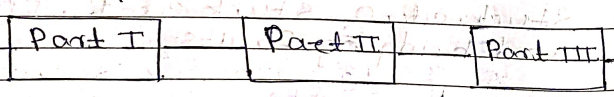
⑨ Hard to reach areas can be tested easily automated tools.

⑩ DFT allows for very high fault Coverage.

⑪ If a chip fails, DFT features can instantly inform the engineer of the exact location of defect

"Adhoc" DFT Technique.

- It is an like Pipeline base structure
- The entire Circuit is divided into Smaller Sub part at a time each Sub part is tested.



- It is divide the larger System into Smallest part, then insert Mux.
- By using the mux it is possible to test dll portion.
- It also includes a Control Signal with the help of this we can access Smaller Part of the Ckt.
- The mux provides the test points

* Guidelines for Adhoc

- Avoid the Combinational flb loop
- Seprates the analog & digitial part of the Ckt.
- Avoid reductant logic ckt.
- Avoid asynchronous logic Ckt.
- Spilt the larger System into Smaller point.
- Reset flf.

* Testability

- Testability is a ability to which indicates how much it is easy to Test Ckt.

① To detects defects easily.
- Helps To find fault like se as broken line durning manufacture

② To test Complex Circuit.

- modern chips are very complicated, Testability makes Testing easiee.

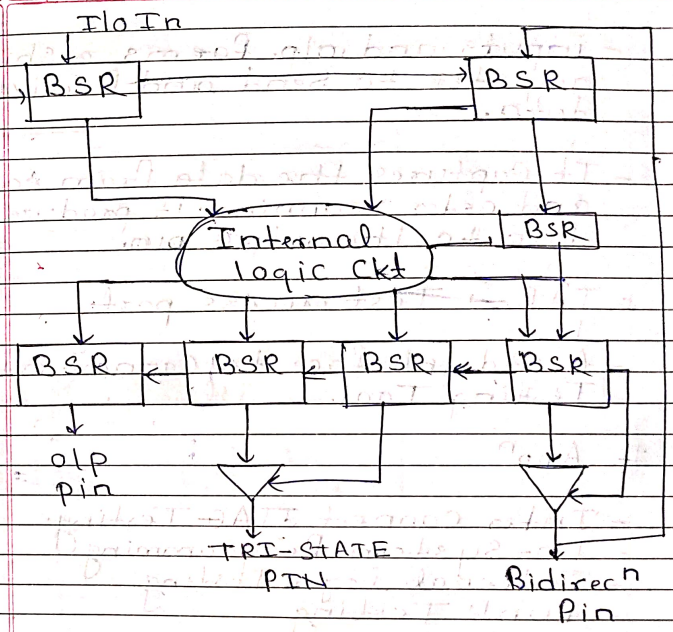
③ To Save the time and Cost.

- Testing becomes faster and cheaper with built in test feature.

⑤ To improve quality & reliabilty

JTAG [Boundary Scan]

- Joint tag test action group.
- We have to test the circuit or we have to test the interal logic of Circuit without making any phy Contact.
- In this method all the flf's are modified into Scan cell, we can Operate it normal it Normal mode & Scan mode.
- JTAG The Boundary Scan Method is a widely used DFT std Method as a part of JTAG.



- In this case of normal ckt, the Circuit is Complicated. It contains millions of Transistors.
- So, in such case, it is very difficult to physically access the particulars point and then test olp at that Point.

o BSR → Boundary Scan Registee - group of Scan Cells Connected to

inputs and o/p forms a shift register to Send and Receive test data.

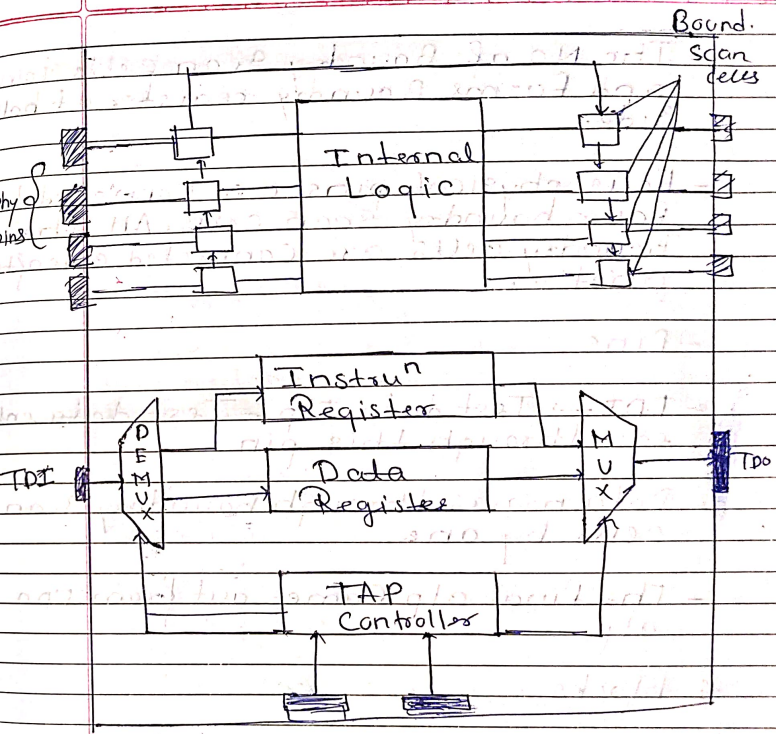
- It Captures the data from each pin and after Scanning it produce the data to the output pin.

o TAP → Test access port.

- A std interface to Connect external Testing Tool.

Appⁿ

- Inter Connect JTAG Testing.
- In-System Peogramming
- Internal logic Testing.
- funⁿ Testing.



- The Central of Block Represents internal logic. There are four Square boxes at the RHS & LHS.

- Each Square box represent the boundary Scan Cell.

- The No of Boundry Scan Cells join and forms Boundry register at both side.

- Four physical pins are connected to four boundry Scan cells. All the boundry cells are connected one after another.

- Pins

- TDI → Test data Ilp – Test data enters through this pin

- Data moves through boundary Scan cells by one.

- The final o/p comes out from TDO -o/p

* blocks

① Instruⁿ Reg

- Stores instruction required for different Test Operation
- works like a Shift Reg.
- Select which Test to run.

② Data Registee

- Holds the data to be applied to the Circuit.
- Sends data to the Mux X then to TDO

③ TAP Controller

- Conteols whole boundey Scan Process.
- Generates Clock and Control Signals for Testing.

Testability Parameters

① Controllability

- Ability to set a node in the Ckt to logic 0 or 1
- means how easily we can Control internal signal using input values
- High Controllability.

② observability

- Ability to see or measure the value of an internal node at the o/p.
- High observability → internal changes quickly appear at o/p

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- low observability ⟶ faults are hidden and difficult to detect.

③ predictability

- Ability to predict the Correct o/p for given inputs.
- Helps Compare expected Vs actual results durning Testing.
- makes fault detection easier.